

Log Analysis & Attack Detection

Introduction

This task focuses on **log analysis and attack detection** using a Security Information and Event Management (SIEM) system within a controlled lab environment. The objective is to simulate real-world cyberattacks, capture system logs, and analyze security events to identify malicious activities. By doing so, the task demonstrates how organizations monitor, detect, and respond to threats in real time. The lab environment consists of three main components: an attacker machine (Kali Linux), a target system (Ubuntu Server), and a SIEM solution responsible for collecting and analyzing logs. A brute-force attack using Hydra is simulated against the target system, while the SIEM platform captures authentication logs and generates alerts for suspicious behavior. These alerts are then analyzed to understand the nature, severity, and impact of the attack, followed by an AI-assisted risk assessment and reporting process.

Although the task recommended the use of Security Onion or the ELK Stack as the SIEM solution, these platforms were not deployed in this implementation. This decision was primarily due to their **large installation size and high system resource requirements**, which can significantly impact system performance and potentially lead to instability or crashes in a virtual lab environment with limited resources. To ensure a stable and efficient setup, **Wazuh** was selected as an alternative SIEM solution.

Wazuh provides a lightweight yet powerful platform for log collection, threat detection, and security monitoring. It offers similar core functionalities such as real-time alerting, log analysis, and dashboard visualization while being more suitable for environments with constrained system resources. By using Wazuh, the task objectives were successfully achieved without compromising system performance, demonstrating a practical and resource-conscious approach to security monitoring.

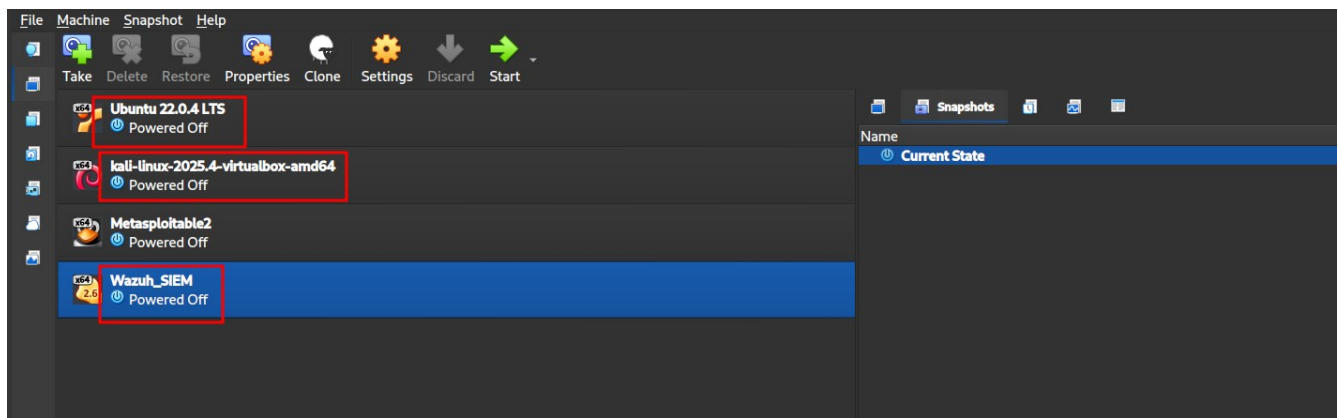
Task Workflow

1. Set up a SIEM (Wazuh)
2. Generate attack traffic (Hydra SSH brute-force)
3. Capture logs
4. Detect and analyze the attack
5. Perform AI risk assessment
6. Produce reports + screenshots

Hands-On

I. Set up Lab Architecture

- i) Kali Linux (Attacker)
- ii) Ubuntu Server (Target)
- iii) Wazuh (SIEM) VM



2. Assign IPs

Kali – 192.168.56.4

Wazuh – 192.168.56.7

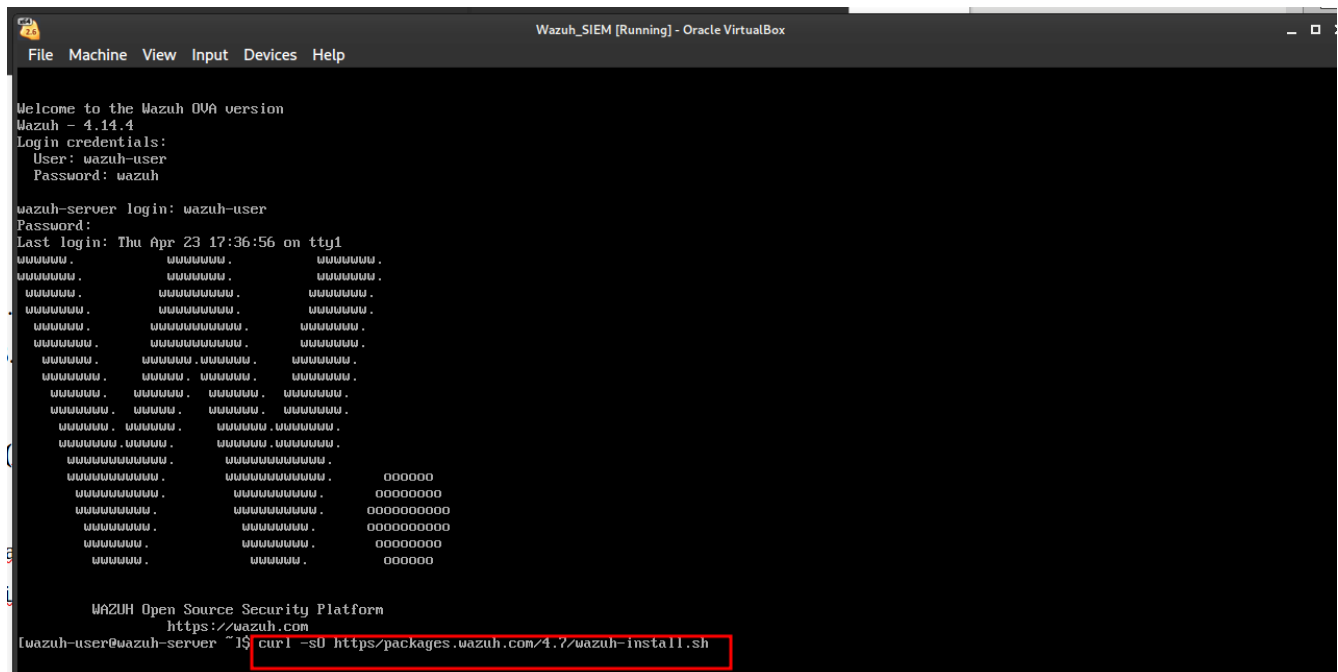
Ubuntu - 192.168.56.8

3. Deploy Wazuh (SIEM)

i) Install Wazuh

```
curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
```

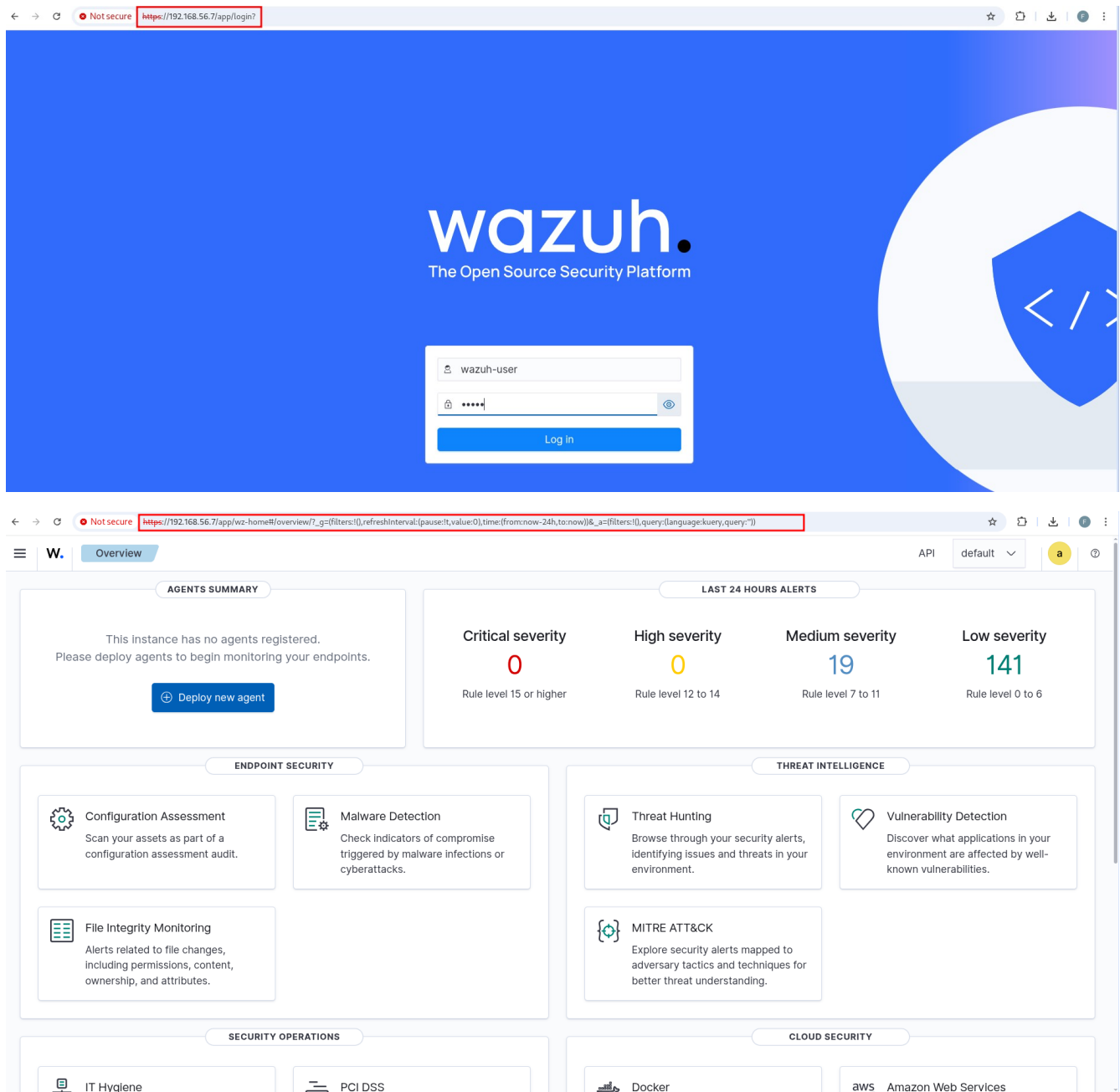
```
sudo bash wazuh-install.sh -a -i
```



4. Access Dashboard

Open browser:

<https://192.168.56.7>



5. Connect Target VM (Ubuntu) to SIEM (Wazuh)

In Ubuntu Terminal

Run:

```
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --no-default-keyring --keyring  
gnupg-ring:/usr/share/keyrings/wazuh.gpg --import && sudo chmod 644  
/usr/share/keyrings/wazuh.gpg
```

```
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable  
main" | sudo tee /etc/apt/sources.list.d/wazuh.list
```

```
sudo apt update
```

```
sudo WAZUH_MANAGER='192.168.56.7' apt install wazuh-agent
```

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable wazuh-agent
```

```
sudo systemctl start wazuh-agent
```

```
cybertribe@cybertribe-VirtualBox:~$ curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH | sudo gpg --no-default-keyring --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --import && sudo chmod 644 /usr/share/keyrings/wazuh.gpg  
gpg: keyring '/usr/share/keyrings/wazuh.gpg' created  
gpg: directory '/root/.gnupg' created  
gpg: /root/.gnupg/trustdb.gpg: trustdb created  
gpg: key 96B3EE5F29111145: public key "Wazuh.com (Wazuh Signing Key) <support@wazuh.com>" imported  
gpg: Total number processed: 1  
gpg: imported: 1  
cybertribe@cybertribe-VirtualBox:~$ echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main" | sudo tee /etc/apt/sources.list.d/wazuh.list  
deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main  
cybertribe@cybertribe-VirtualBox:~$ sudo apt update  
Hit:1 http://ci.archive.ubuntu.com/ubuntu jammy InRelease  
Hit:2 http://ci.archive.ubuntu.com/ubuntu jammy-updates InRelease  
Hit:3 http://ci.archive.ubuntu.com/ubuntu jammy-backports InRelease  
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease  
Get:5 https://packages.wazuh.com/4.x/apt stable InRelease [17.3 kB]  
Get:6 https://packages.wazuh.com/4.x/apt stable/main amd64 Packages [51.9 kB]
```

```
cybertribe@cybertribe-VirtualBox:~$ sudo WAZUH_MANAGER='192.168.56.7' apt install wazuh-agent  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following NEW packages will be installed:  
  wazuh-agent  
0 upgraded, 1 newly installed, 0 to remove and 49 not upgraded.  
Need to get 13.2 MB of archives.  
After this operation, 48.9 MB of additional disk space will be used.  
Get:1 https://packages.wazuh.com/4.x/apt stable/main amd64 wazuh-agent amd64 4.14.5-1 [13.2 MB]  
Fetched 13.2 MB in 4min 28s (49.2 kB/s)  
Preconfiguring packages ...  
Selecting previously unselected package wazuh-agent.  
(Reading database ... 208794 files and directories currently installed.)  
Preparing to unpack .../wazuh-agent_4.14.5-1_amd64.deb ...  
Unpacking wazuh-agent (4.14.5-1) ...  
Setting up wazuh-agent (4.14.5-1) ...  
cybertribe@cybertribe-VirtualBox:~$ sudo systemctl daemon-reload  
sudo systemctl enable wazuh-agent  
sudo systemctl start wazuh-agent  
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service  
→ /lib/systemd/system/wazuh-agent.service.
```

6. Enable SSH on Target (Ubuntu)

```
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh
```

```
cybertribe@cybertribe-VirtualBox:~$ sudo apt install openssh-server -y
[sudo] password for cybertribe:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
```

```
cybertribe@cybertribe-VirtualBox:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/sy
stemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
cybertribe@cybertribe-VirtualBox:~$ sudo systemctl start ssh
cybertribe@cybertribe-VirtualBox:~$
```

7. Run Attack from Kali using Hydra

```
hydra -l root -P /usr/share/wordlists/rockyou.txt.gz ssh://192.168.56.8
```

```
(kali) kali) ~
$ hydra -l root -P /usr/share/wordlists/rockyou.txt.gz ssh://192.168.56.8
Hydra v9.6 (C) 2023 by van hauser/thc & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-24 17:56:32
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.56.8:22/
```

8. Detection in Wazuh

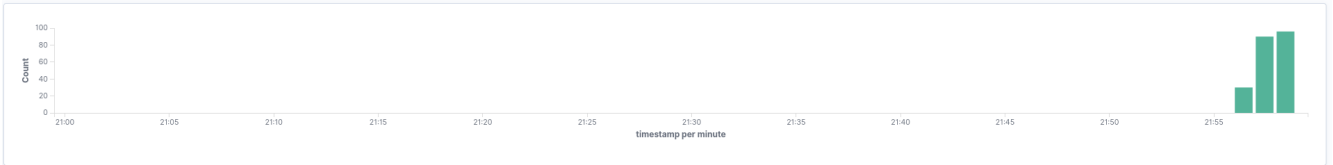
View Alerts

Wazuh Dashboard → Security Events → Filter: ssh OR authentication failed

Dashboard Events Explore agent

ssh DQL Last 1 hour Show dates Refresh

manager.name: wazuh-server Add filter



216 hits
Apr 24, 2026 @ 20:59:28.510 - Apr 24, 2026 @ 21:59:28.510

Export Formatted Reset view 641 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Apr 24, 2026 @ 21:58:53.166	cybertribe-VirtualBox	PAM: Multiple failed logins in a small period of time.	10	5551
Apr 24, 2026 @ 21:58:53.165	cybertribe-VirtualBox	PAM: User login failed.	5	5503
Apr 24, 2026 @ 21:58:51.369	cybertribe-VirtualBox	PAM: User login failed.	5	5503
Apr 24, 2026 @ 21:58:51.369	cybertribe-VirtualBox	PAM: User login failed.	5	5503
Apr 24, 2026 @ 21:58:51.369	cybertribe-VirtualBox	PAM: User login failed.	5	5503
Apr 24, 2026 @ 21:58:51.366	cybertribe-VirtualBox	PAM: User login failed.	5	5503
Apr 24, 2026 @ 21:58:51.324	cybertribe-VirtualBox	PAM: User login failed.	5	5503
Apr 24, 2026 @ 21:58:51.309	cybertribe-VirtualBox	PAM: User login failed.	5	5503